

JEE MAIN REVISION MASTER: MOTION IN A PLANE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Projectile Motion: Range, Maximum Height & Flight Time	8 PYQs	2D Kinematics
Standard	Relative Velocity in 2D (Rain-Man & River-Boat)	1 PYQs	Relative Motion

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Projectile Motion: Range, Maximum Height & Flight Time	$R = \frac{u^2 \sin 2\theta}{g}$, $H = \frac{u^2 \sin^2 \theta}{2g}$, $R_{\max} = \frac{u^2}{g}$ (at $\theta = 45^\circ$)
Relative Velocity in 2D (Rain-Man & River-Boat)	$t_{\min} = \frac{d}{v_b}$, $\text{Shortest Path: } \sin \theta = \frac{v_r}{v_b}$, $\vec{v}_{\text{rain, man}} = \vec{v}_r - \vec{v}_m$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Projectile Motion: Range, Maximum Height & Flight Time (High Yield — 8 PYQs)

Core Principle: 2D parabolic projectile motion under uniform downward gravity. Horizontal range $R = (u^2 \sin 2\theta / g)$, maximum height $H = (u^2 \sin^2 \theta / 2g)$, flight time $T = (2u \sin \theta / g)$. Horizontal velocity $u \cos \theta$ remains invariant throughout.

■ **Caution / JEE Pitfall:** Horizontal range is identical for complementary angles θ and $(90^\circ - \theta)$, but maximum heights and flight times are different!

■ **Speed Trick:** Relation between Range and Max Height: $R = 4H \cot \theta$. At $\theta = 45^\circ$, $R = 4H$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizo..." [Ans: B]
- [2025 Shift] Q#2 (MCQ): "In the projectile motion of a particle on a level ground, which of the following remains constant with reference to time and position?..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "Two objects are projected at an angle θ° and $(90-\theta^\circ)$, to the horizontal with the same speed. The ratio of their..." [Ans: N/A]

3.2 Relative Velocity in 2D (Rain-Man & River-Boat) (Standard Yield — 1 PYQs)

Core Principle: Relative velocity $\text{vec}(v)_{AB} = \text{vec}(v)_A - \text{vec}(v)_B$. Crossing a river in shortest time requires heading directly perpendicular to the bank ($t_{\min} = d/v_b$). Crossing along the shortest path (zero drift) requires swimming upstream at $\sin\theta = v_r/v_b$.

■ **Caution / JEE Pitfall:** To protect from vertically falling rain when walking forward, hold the umbrella tilted FORWARD in the direction of motion at $\tan\theta = v_m/v_r$.

■ **Speed Trick:** Shortest path across a river is only possible if swimmer speed $v_b > v_r$. If $v_b < v_r$, zero drift is impossible.

Representative Source PYQs:

- **[2020 Shift] Q#4 (MCQ):** "Rain is falling vertically with a speed of 12ms^{-1} . A woman rides a bicycles with a speed of 12ms^{-1} in east to west direction. Wha..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Two bodies are projected with the same velocity. If one is projected at an angle of 30° and the other at 45° to the horizontal, then the ratio of maximum heights attained is

- (A) 3:1
- (B) 1:2 **[Correct]**
- (C) 4:1
- (D) 1:3

Correct Answer: B

Solution Analysis: For a projectile, the maximum height is given by $H = \frac{u^2 \sin^2\theta}{2g}$. Here both bodies are projected with the same velocity, so u is the same for both. Also, g is the same. So maximum height depends only on $\sin^2\theta$. For the first body, $\theta = 30^\circ$, $H_1 = \frac{u^2 \sin^2 30^\circ}{2g}$. For the second body, $\theta = 45^\circ$, $H_2 = \frac{u^2 \sin^2 45^\circ}{2g}$. So, $\frac{H_1}{H_2} = \frac{\sin^2 30^\circ}{\sin^2 45^\circ}$. We know, $\sin 30^\circ = (1/2)$ and $\sin 45^\circ = \frac{1}{\sqrt{2}}$. So, $\frac{H_1}{H_2} = \frac{(1/2)^2}{(1/\sqrt{2})^2} = \frac{1/4}{1/2} = \frac{1}{2}$.

Benchmark Problem 2: KCET 2020 (Q#4)

Rain is falling vertically with a speed of 12ms^{-1} . A woman rides a bicycles with a speed of 12ms^{-1} in east to west direction. What is the direction in which she should hold her umbrella?

- (A) 30° , towards east
- (B) 45° , towards east
- (C) 30° , towards west
- (D) 45° , towards west

Correct Answer: None

Solution Analysis: The given situation is shown below. Here, velocity of rain, $v_r = 12\text{ms}^{-1}$ and velocity of woman, $v_w = 12\text{ms}^{-1}$. The relative velocity of rain w.r.t. woman is $v_{nw} = \sqrt{v_w^2 + v_r^2} = \sqrt{(12)^2 + (12)^2} = 12\sqrt{2}\text{ms}^{-1}$. The direction in which the woman should hold her umbrella is $\sin\theta = \frac{v_w}{v_{nw}} = \frac{12}{12\sqrt{2}} \Rightarrow \theta = \sin^{-1}(\frac{1}{\sqrt{2}}) = 45^\circ$, towards east.